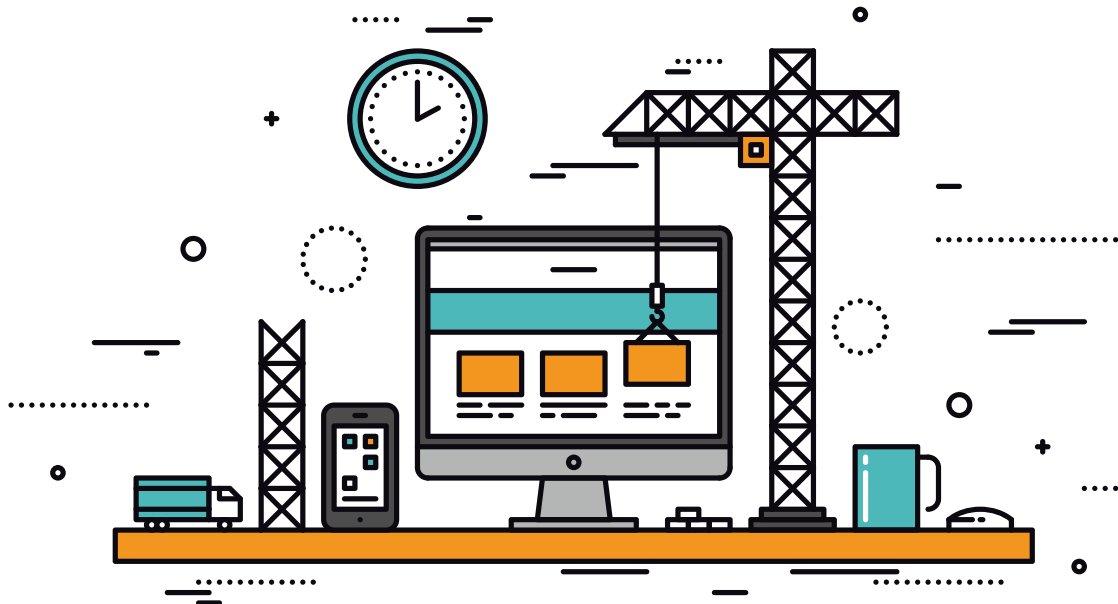


Build a Web Application on Firebase

Firebase is a platform backed by Google for building, growing and deploying web apps, mobile apps and games. This article illustrates developing a complete single-page web application on Firebase using its services, namely, Firestore, authentication and hosting. The readers can follow the steps given here and develop a web application with ease.



This article will handhold you to develop a simple web application to maintain a journal of events.

Requirements

To start with, let us specify a clear requirement. Requirement specification is one of the most important aspects of the development process. Here is the list of requirements.

- The journal should be accessible only via login.
- The login credentials are predefined by the system administrator, and consist of email and password.
- A logged-in user should be able to add a new entry into the journal.
- A logged-in user should be able to view the entries in the journal.
- A logged-in user should be able to sort the entries chronologically.
- Each entry must auto-fill the details like date of entry and author.

Architecture

The application is designed around the Firebase cloud platform. The platform offers services, scale, SDKs, tools and well-maintained documentation. The following services are used for this application.

- **Firestore:** The real-time database service for storing the journal entries.
- **Hosting:** The service to host the web application.
- **Authentication:** The service to offer login-based access to the application.
- **Firebase SDK:** The NPM module that offers commands and communication API. The JS client uses this API to interact with the Firestore and authentication services at runtime. The architecture is depicted in Figure 1.

So what do you need to get started with building the application? Here is the list of basic prerequisites.

- An account with Google (like a Gmail account)
- Node 10+ distribution on the development machine